ANU Fitness: Full-Stack Smart Fitness & Nutrition Tracking Platform

A comprehensive web-based fitness management system combining workout planning, meal tracking, and analytics

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Problem Statement: Fragmented Fitness Tracking Creates User Confusion and Inconsistency

Multiple Apps

Users juggle multiple apps for workouts, calories, and progress tracking

Fragmented Management

No centralized platform for complete fitness management

Data Loss Risk

Manual notes and separate tools lead to time waste and data loss

Limited Insights

Lack of integrated analytics prevents meaningful progress insights

Solution Purpose: Unified Fitness Dashboard

Centralized Platform

Centralizes workout planning, meal logging, and analytics in one platform

Personalized Goals

Provides personalized goals and customizable fitness targets

Data-Driven Insights

Enables data-driven fitness decisions through visual analytics

Intuitive Interface

Simplifies daily fitness management with intuitive user flow

Secure Authentication: JWT-Based Login System with bcrypt Password Protection

Local Authentication

User registration and secure login mechanism with credential validation and session management to ensure authorized access control

bcrypt Password Protection

Passwords hashed using bcrypt algorithm to meet security compliance standards and protect user credentials from unauthorized exposure

JWT Token Security

JWT tokens maintain secure sessions and protect routes by enabling stateless authentication and controlling access to protected resources

Data Isolation

User-specific data isolation mechanisms prevent unauthorized access and ensure confidentiality by restricting data visibility to authenticated users

Smart Dashboard: Real-Time Fitness Overview

Central Control Panel

Monitor your daily calorie consumption against your personalized target goals in real-time with intuitive visual indicators

Workout Summaries

Review recent workout session details and performance highlights including duration, intensity, and calories burned

Weekly Progress Trends

Visualize your activity patterns throughout the week with interactive charts showing consistency and improvement over time

Quick Action Buttons

Log new workouts and meal entries instantly with convenient one-tap access to core logging features

Comprehensive Workout Library: 100+ Exercises Organized by Muscle Groups

Filterable by Muscle Group

Chest, Back, Legs, Shoulders, Arms, Core

Search by Difficulty and Equipment

Search by difficulty level and equipment type

Complete Exercise Details

Each exercise includes target muscle, equipment needed, and instructions

Structured Workout Planning

Enables structured workout planning based on training goals

For All Skill Levels

Simplifies exercise selection for beginners and advanced users

Detailed Workout Logging: Session-Based Tracking with Sets, Reps, and Weight Data

Create Named Workout Sessions

Set up personalized workout sessions with custom names such as Leg Day or Push Day to organize your training regimen effectively

Add Multiple Exercises

Include complete performance data for each exercise within your sessions to track detailed progress and maintain comprehensive records

Record Performance Data

Log sets, reps, weight used, duration, and personal notes to capture every aspect of your workout for detailed analysis

Maintain Historical Records

Keep historical workout records accessible for progress tracking and long-term performance analysis across your fitness journey

Precision Food Tracker: Meal-by-Meal Calorie and Macro Monitoring

Categorized Meal Entry

Organize meals by meal type—Breakfast, Lunch, Dinner, and Snacks—for clear, structured food logging and easy meal-by-meal tracking throughout the day.

Track Detailed Nutrition

Log food name, quantity, calories, protein, carbs, and fats for each item. Capture comprehensive nutritional data to support precise dietary analysis and goal achievement.

Automatic Daily Calculations

Get instant daily total calculations for all nutrition metrics. View your complete nutritional overview at a glance for real-time monitoring and informed dietary decisions.

Date-Based Organization

Access historical meal records organized by date for trend analysis and progress tracking. Align past eating patterns with dietary goals and monitor long-term nutrition progress.

Visual Analytics Dashboard: Transforming Raw Data into Actionable Insights

Daily Nutrition Breakdown

Macro distribution charts providing detailed insights into protein, carbohydrates, and fats consumption throughout the day

Weekly Calorie Trend

Visualization showing consistency patterns across the week with clear trend analysis and daily variance tracking

Workout Frequency Analysis

Comprehensive tracking of exercise sessions and total sets completed, revealing workout intensity and consistency patterns

Performance Overview

Comparative analysis of goals versus actual results, highlighting achievements and areas for improvement

Personalized Profile System: Custom Goals for Fat Loss, Maintenance, or Muscle Gain

User-Configurable Fitness Goals

Set custom fitness objectives and personalized calorie targets aligned with your fat loss, maintenance, or muscle gain goals.

Customizable Macro Targets

Define your ideal macronutrient distribution including protein, carbohydrates, and fats to match your dietary preferences and fitness needs.

Weight Tracking and Display

Monitor current weight metrics and customize your display name settings for a personalized user experience throughout the application.

Goal-Based Recommendations

Receive adaptive dashboard calculations and personalized recommendations that automatically adjust based on your selected fitness goals.

Modern Full-Stack Architecture: React + TypeScript Frontend with Node.js + Express Backend

Frontend

React with TypeScript for type safety, Vite for fast development, Tailwind CSS for styling

Backend

Node.js and Express REST API with TypeScript

Data Layer

SQLite database managed through Prisma ORM

Authentication

JWT tokens and bcrypt password hashing

Validation

Zod for input sanitization and data integrity

Robust Database Design: 7-Table Schema Supporting Complex Fitness Data Relationships

User Table

Stores user profiles, fitness goals, and authentication data for comprehensive account management and personalization

Workout Table

Contains exercise library with detailed muscle group targeting and exercise instructions for guided training

MealEntry & MealItem

Hierarchical structure enabling detailed food tracking with meal entries linked to individual food items

WorkoutSession & Exercise

Detailed workout session logging with linked exercise records to track complete training history and performance

WorkoutSet Table

Tracks sets, repetitions, and weight data for each exercise to monitor progressive strength gains and training load

Relational Structure

Complex relationship design enables comprehensive analytics, historical tracking, and performance trend analysis

Project Differentiators: Beyond Basic Fitness Apps with Complete Ecosystem

Comprehensive Content Management

Not just workout content - includes logging, tracking, and analytics for complete health oversight

Full CRUD Operations

Complete Create, Read, Update, Delete operations with user-specific data isolation for security

Real Database Persistence

Integrated real database persistence versus temporary storage, ensuring data durability

Integrated Macro Tracking

Comprehensive macro tracking alongside calorie counting for detailed nutritional analysis

Unified Platform

Combined workout planning and performance logging in single platform demonstrating full-stack expertise

2) Working application

Web app working online: https://client-phi-three-45.vercel.app

GitHub Link: https://github.com/AnnapoornaN/ANU-Fitness.git

Thank You for Your Attention

ANU Fitness demonstrates complete full-stack development expertise: React + TypeScript frontend, Node.js + Express backend, SQLite database, secure authentication, REST APIs, and analytics dashboard. A production-ready fitness management system designed to simplify personal health tracking through a unified, user-friendly platform.